

FUTURE
UNIVERSITY IN EGYPT
جـــامـعـــة المســتقبـــل

Ahmad Jehad Kobah 20225366

Mohamed Eslam 20224285

Sofian Hashmi Shbili 20220658

Department of Computer Science

Semester: Spring 2025

Course: Computer Graphics (Sp25 - DMT 323 & DM341 - Pr)

Documentation Game Pac Man

Content

1. Introduction 2
2. Game Objective 2
3. Development Tools 2
Languages: 2
Platform: 2
4. Functionality 3
Game Mechanics: 3
Scoring System: 3
Collision Detection: 3
Movement: 3
5. Design Layout 4
Maze Structure: 4
UI Elements: 4
Game Over Screen: 4
Visual Effects: 4
6. Future Improvements 5
Ghost Behavior: 5
Sound Effects: 5
Level Progression: 5
High Score System: 5
Mobile Support: 5
7. Conclusion 5

Pac-Man Game Development Report

1. Introduction

This report provides an overview of the **Pac-Man** game developed using **HTML, CSS, and JavaScript**. The game is a simplified recreation of the classic arcade game, featuring **Pac-Man, ghosts, pellets, and a maze layout**. The objective is to navigate the maze, collect pellets, avoid ghosts, and achieve the highest score possible.

---

2. Game Objective

- The player controls **Pac-Man**, guiding him through a maze to **consume all pellets**.
- **Avoid ghosts**—colliding with them reduces lives.
- **Power pellets** (larger dots) temporarily allow Pac-Man to eat ghosts.
- The game ends when:
    - All pellets are collected (**win**).
    - All lives are lost (**game over**).

---

3. Development Tools

Languages:

- **HTML5** (Structure)
- **JavaScript** (Game Logic & Mechanics)

Platform:

- **Web Browser** (Chrome, Firefox, Edge, etc.)
- **No external libraries**—pure vanilla JS for performance and simplicity.

---

4. Functionality

Game Mechanics:

- **Player Movement:**
    - Controlled via **arrow keys (↑, ↓, ←, →)**.
    - Smooth movement with **collision detection against walls**.
- **Ghost AI:**
    - Random movement with **basic pathfinding** (avoids walls).
    - Changes direction periodically.
- **Pellet System:**
    - **Regular pellets** (10 points).
    - **Power pellets** (50 points, enable ghost-eating).

Scoring System:

- **Regular pellet:** +10 points.
- **Power pellet:** +50 points.
- **Score displayed in real-time** at the bottom-right.

Collision Detection:

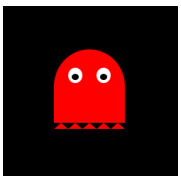
- **Wall Collision:** Prevents Pac-Man and ghosts from passing through walls.
- **Ghost Collision:**
    - If Pac-Man touches a ghost:
        - **Lose a life.**
        - **Brief invincibility period** (blinking effect).
        - **Reset positions** (Pac-Man and ghosts return to starting points).

Movement:

- **Pac-Man:**
    - Moves at **tileSize / 16 speed** (smooth but challenging).
    - Mouth animation (opens/closes while moving).



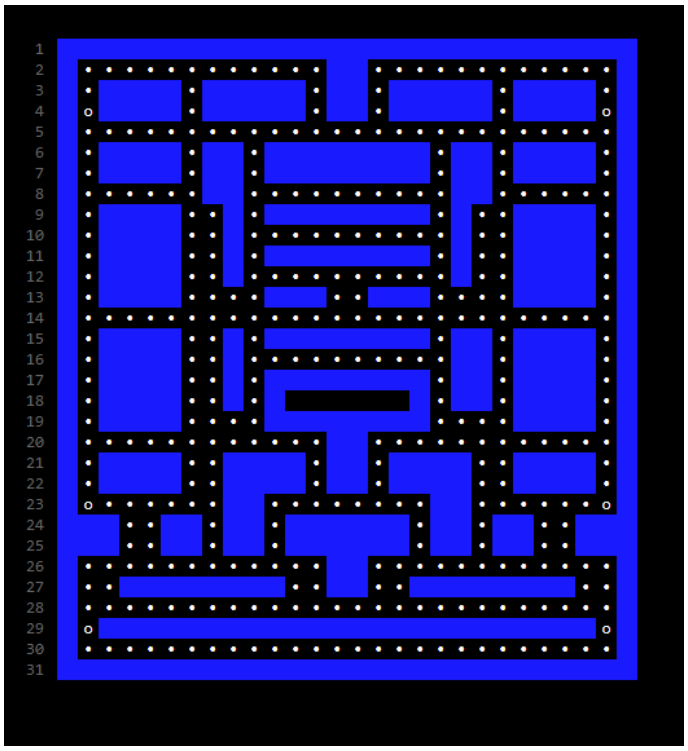
- **Ghosts:**
    - Move at **tileSize / 12 speed** (slightly faster than Pac-Man).
    - Random direction changes with **cooldown delays**.



5. Design Layout

Maze Structure:

- **28x31 grid** (448x496 pixels).
- **Walls (blue), pellets (white), power pellets (larger white dots).**
- **Tunnels** (left/right wrap-around).
- **Ghost house** (center area where ghosts spawn).



UI Elements:

- **Lives Counter (Bottom-Left):** Shows remaining lives (max 3).
- **Score Counter (Bottom-Right):** Updates in real-time.
- **Restart Button:** Appears after **game over/win**.

Game Over Screen:

- **"GAME OVER"** displayed with final score.
- **"YOU WIN!"** if all pellets are collected.
- **Semi-transparent overlay** for visibility.

Visual Effects:

- **Pac-Man blinking** when invincible after losing a life.
- **Smooth animations** for movement and pellet collection.

6. Future Improvements

Ghost Behavior:

- Different movement patterns (e.g., chase, scatter, frightened modes).

Sound Effects:

- Eating pellets, ghost collisions, background music.

Level Progression:

- Increasing difficulty with faster ghosts.

High Score System:

- Local storage to save best scores.

Mobile Support:

- Touch controls for smartphones/tablets.

---

7. Conclusion

This **Pac-Man** clone successfully implements core gameplay mechanics while maintaining simplicity. The use of **pure JavaScript** ensures smooth performance, and the **responsive UI** enhances player experience. Future updates could expand features like **advanced AI, power-ups, and multiplayer modes**.

This project demonstrates **strong fundamentals in game development** using web technologies and serves as a foundation for more complex games.